

Yinting (Justin) Huang

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Education

University of Cambridge, United Kingdom

Oct 2026 – Present

PhD in Computer Science with CASCADE Studentship

Imperial College London, United Kingdom

Oct 2020 – Jul 2024

MEng Electrical and Electronic Engineering

First-Class Honours

Dean's List for Academic Excellence (2021, 2022)

Skills

- Bash, C++, Python, SystemVerilog, MATLAB, gem5, ChampSim, PyTorch, RISC-V, Linux, Vivado, Quartus, Cadence, GDB, Git, FPGA

Publications and Research Outputs

- **Proba: A High-Performance, Low-Traffic Probabilistic Spatial Memory Streaming Prefetcher**, in the 35th International Conference on Parallel Architectures and Compilation Techniques (PACT 2026)
- **Caerus: Prefetching by Thinking Globally and Acting Locally**, in the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026)
- **ITERA-LLM: Boosting Sub-8-Bit LLM Inference via Iterative Tensor Decomposition**, in the 33rd International Symposium on Field-Programmable Custom Computing Machines (FCCM 2025) [[Paper](#)]
- **Integration of RISC-V Page Table Walk in gem5 SE Mode**, in the Proceedings of the 16th Workshop on Rapid Simulation and Performance Evaluation for Design (RAPIDO 2024) [[Paper](#)]

Industrial Placement

CPU Architecture Research with Dr. Sam Ainsworth

Sept 2024 - Sept 2026

- Conducted research on state-of-the-art **hardware prefetching** techniques
- Designed, modelled, and evaluated different types of hardware prefetchers in the **gem5 simulator**
- Optimised the temporal prefetcher **Triangel** by introducing a **delta representation** to capture **spatio-temporal access streams** with near-zero metadata overhead
- Demonstrated that **Triangel-Delta** delivers a **3% geometric-mean speedup** and a **1.2% DRAM-traffic reduction** over baseline Triangel on a set of memory-intensive SPECspeed 2017 benchmarks
- Designed **Proba**, a novel Spatial Memory Streaming (SMS) prefetcher that employs **probabilistic pattern updates**, PC-grained pattern aggregation, and multi-level accuracy feedback mechanisms for fine-grained accuracy-coverage control (**patent under review**)
- Demonstrated that Proba achieves a **2.3% geometric-mean speedup** and a **7.0% DRAM-traffic reduction** over SMS, and a **1.1% speedup** and **15.9% DRAM-traffic reduction** over DSPatch, across the full SPECspeed 2017 benchmarks with competitive storage cost
- Contributed to the design of the offset prefetcher **Caerus** using **PC-local filtering** over **multiple globally derived offsets**

CPU Architecture Internship

April 2023 - Sept 2023

- Automated benchmarking and performance analysis targeting RISC-V and AArch64
- Implemented **RISC-V Sv39 virtual address translation and page walk** in gem5 SE mode

MEng Thesis

ITERA-LLM:

Jun 2024 – Jan 2025

Boosting Sub-8-Bit LLM Inference via Iterative Tensor Decomposition

- Designed an iterative LLM compression algorithm based on **quantisation, SVD-based low-rank decomposition, and sensitivity-based rank allocation**
- Developed a **hardware-agnostic layer-wise optimisation algorithm** by tuning the decomposition rank of each layer based on a sensitivity estimate
- Implemented a library of **compression-aware** and **parametrisable** sparse MatMul engines to generate an accelerator that optimises hardware utilisation and computation latency
- Developed analytical models to estimate the performance and resource consumption of compressed LLMs on FPGA platforms, supporting **automated design space exploration** to identify optimal design points balancing accuracy and latency
- Achieved **linear layer latency reductions** ranging from **12.1%** to **41.1%**, compared to state-of-the-art quantisation-only MatMul baseline with comparable model accuracy

Projects

Function Accelerator Based on Cyclone V FPGA

Semester Long Project

- Designed a function accelerator to perform **floating-point subtraction, multiplication, vector summation, and trigonometric functions**, achieving a **600×** performance improvement
- Configured the **NiosII soft processor core, external memories and custom instruction interfaces** to integrate with the dedicated accelerator
- Implemented custom **SystemVerilog** modules including **adder trees, CORDIC processors, floating-to-fixed-point converters, custom instruction interfaces, and pipeline controllers** to minimise computation latency and maximise throughput
- Conducted comparative analysis on FPGA resources, latency, accuracy and application size

Pipelined RISC-V CPU Model

Semester Long Project

- Developed a **SystemVerilog** CPU model featuring the **RV32IM ISA** and interrupt handling
- Verified the CPU model using **RISC-V assembly, C++ testbench, Verilator, and GTKWave**
- Implemented a **multi-cycle multiplier and divider microarchitecture**
- Integrated a **virtual RAM module** with a Wishbone B4 classic interface
- Designed a **set-associative, least-recently-used, write-back** cache controller

References and research outputs are available upon request.